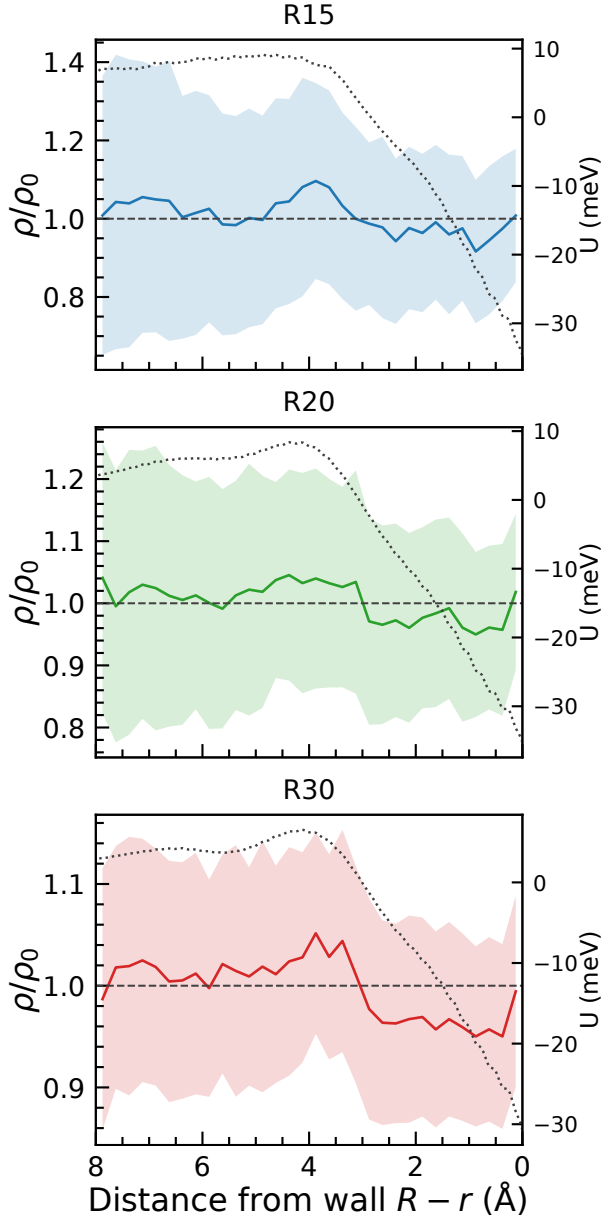Density fluctuations and mean-field potential